Part 4: Where Are We, and Which Way Are We Going? Onboard Orientation Software

Sakarias Frette[1]

[1] *AST2000 Rerun, University of Oslo*

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A spacecraft must find its own state from its instruments. We build software that does this right after launch.

- **Pointing.** We match a camera picture against 360 reference pictures that we project stereographically from a map of the full sky. The tested pointing angles are accurate to 0.23°, and remain within the API's 2° tolerance up to pixel noise of $\sigma = 16$.
- **Velocity.** We combine the Doppler shifts of two reference stars, which gives 41.15 km/s relative to the star.
- **Position.** We trilaterate from the distances to all planets and the star, by a linear least-squares solution that needs no initial guess.

The software agrees with our own launch prediction to 1.1 km and 7.4 m/s, and `ast2000tools` verified it (passed). Along the way we found that the API defines the Doppler measurement differently from the textbook. With our star geometry, the textbook reading would be 27% off.

I. INTRODUCTION

After launch, the spacecraft knows nothing about its state except what its instruments tell it [1]. It has three instruments:

- a camera, which gives the pointing angle (Section II);
- a spectrograph that measures Doppler shifts, which gives the velocity (Section III);
- a radar that measures distances, which gives the position (Section IV).

The function `orient(mission)` combines all three. Part 5 calls it again and again during the journey. Solar system seed: 42716.

II. PICTURES OF THE SKY

A. Stereographic projection

The sky map is a HEALPix map [2] with one colour per direction (θ, ϕ). A camera with a 70° field of view sees a flat picture: a stereographic projection of a patch of the sphere. For a picture centred at $\theta_0 = 90°$, ϕ_0, a point (X, Y) in the picture plane maps back to

$$\theta = \frac{\pi}{2} - \arcsin \frac{Y \sin\beta}{\rho}, \quad \phi = \phi_0 + \arctan \frac{X \sin\beta}{\rho \cos\beta}, \quad (1)$$

with $\rho = \sqrt{X^2 + Y^2}$ and $\beta = 2\arctan(\rho/2)$. The API looks up one pixel per call. 360 full pictures would take 10^8 calls, so we ported its HEALPix formula to NumPy. Our lookup agrees with the API's in all 20 000 tested directions (0 mismatches). Our picture for $\phi_0 = 0$ equals the course's sample picture up to its added noise, which has a standard deviation of 4.0 (Fig. 1).

B. Finding the pointing angle

We compare a picture with reference pictures for $\phi_0 = 0, 1, \ldots, 359°$ by the mean squared colour difference. We take the best whole degree and refine it with a parabola through its neighbours. The key step is to compare the pictures after averaging 8×8 pixel blocks. Raw pixels are dominated by sharp stars, and a picture rotated by only half a degree already differs as much as a random one (Fig. 3, right). A library of whole degrees would then have no valley to slide into. The blurred pictures keep the bright galactic band, which changes smoothly with angle, and so give one wide minimum.

C. Results

The software finds 0.00° for the sample picture, 20.00° for the second sample, and 171.00° for the spacecraft's own picture (Fig. 2). On 20 noisy pictures at random angles the worst error is 0.23°. Figure 3 shows the noise margin. The pointing stays within 2° up to pixel noise of $\sigma = 16$, far above the noise of 4.0 in real pictures. It fails only once the noise is comparable to the galactic band's contrast.

III. VELOCITY FROM DOPPLER SHIFTS

A. Method

A source receding at speed $v_r \ll c$ shifts a spectral line by $\Delta\lambda/\lambda_0 = v_r/c$, with $\lambda_0 = 656.3$ nm (Hα). Two reference stars, at $\phi = 45.0°$ and 293.4°, recede from the Sun at -9119.3 and +7995.0 m/s. The difference between the shifts seen from the Sun and from the spacecraft gives the spacecraft's speed v_i towards each star.

Turning (v_1, v_2) into (v_x, v_y) has two readings.

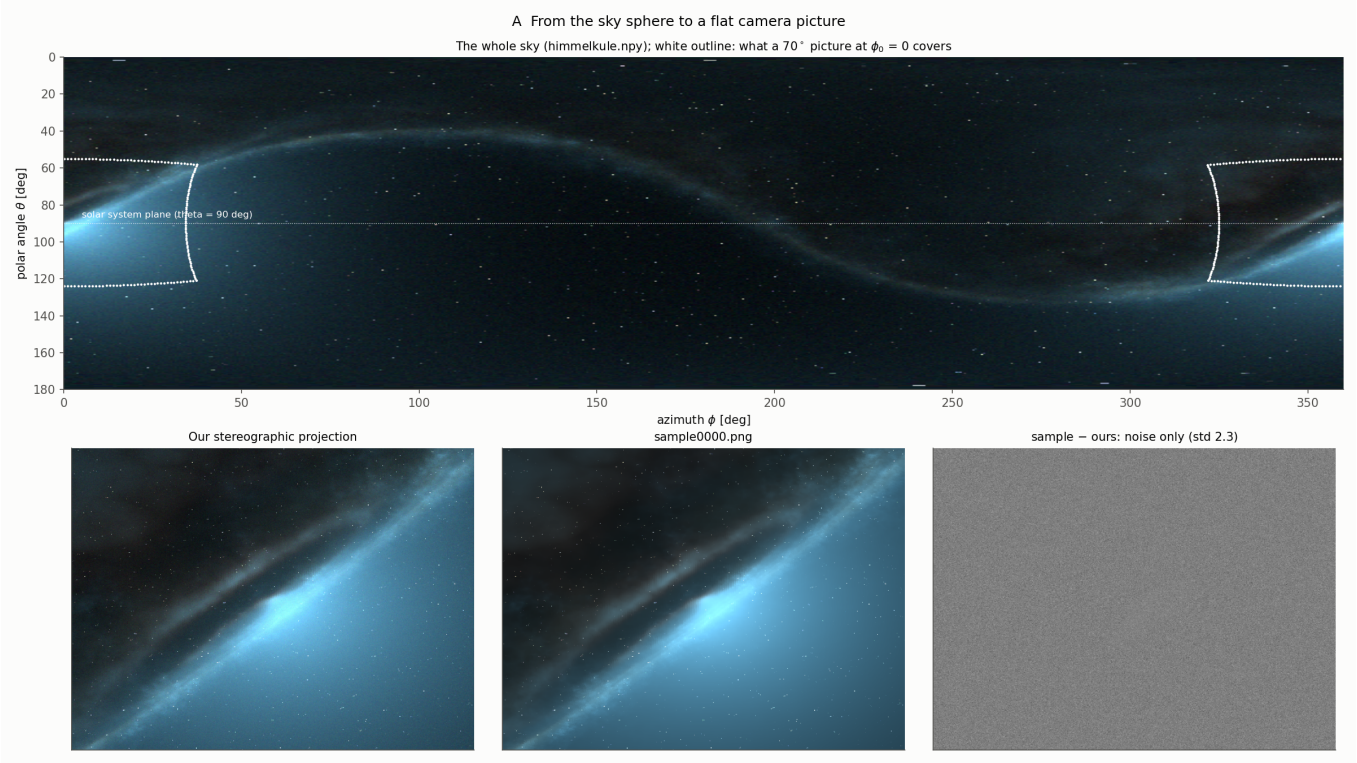


FIG. 1. The whole sky (top) with the area a 70° picture covers, our projection, the sample picture, and their difference.

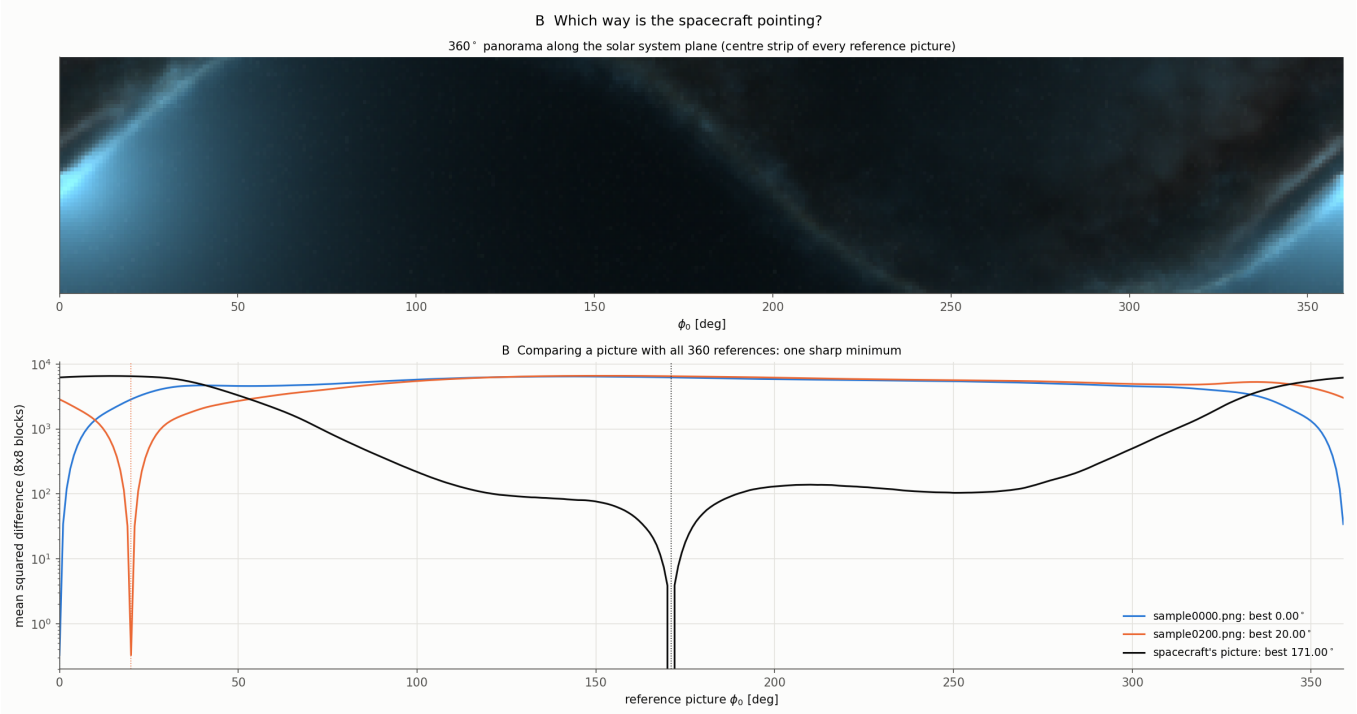


FIG. 2. A panorama of the reference pictures along the ecliptic, and the match score against angle for each test picture.

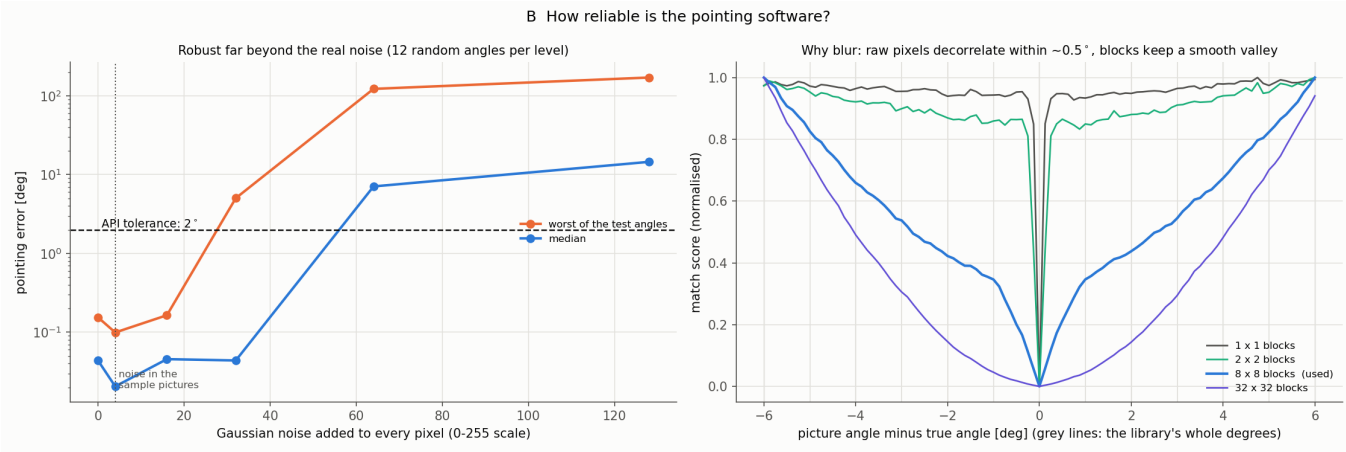


FIG. 3. Left: pointing error against added pixel noise. Right: match score near the true angle for different block sizes; blurring is what makes a whole-degree library work.

- The textbook projection, $v_i = \mathbf{v} \cdot \hat{\mathbf{u}}_i$.
- The basis expansion $\mathbf{v} = v_1\hat{\mathbf{u}}_1 + v_2\hat{\mathbf{u}}_2$, which is how `ast2000tools` generates its data, as we found by reading its source.

The two agree only for perpendicular stars. Ours are 111.6° apart, and the textbook reading would be 27% off (Fig. 4).

B. Results

Right after launch the spacecraft moves at $(2.4534, 8.3257)\,\mathrm{AU/yr}$, which is $41.15\,\mathrm{km/s}$. Self-tests check three cases:

- the Sun's own shifts give $\mathbf{v} = 0$;
- random velocities survive a round trip through simulated shifts;
- both readings agree for perpendicular stars.

The error budget (Fig. 5) is simple. Every $10^{-6}\,\mathrm{nm}$ of wavelength error is $0.46\,\mathrm{m/s}$ along one star direction. With the API's basis expansion, the geometry of the stars does not amplify this. To stay within the API's tolerance of 1% of $|\mathbf{v}|$ ($411\,\mathrm{m/s}$), each shift must be measured to better than $6.4 \times 10^{-4}\,\mathrm{nm}$.

IV. POSITION BY TRILATERATION

A. Method

At time t the planets' positions $\mathbf{p}_i$ are known exactly from Kepler's equation (Part 2), and the star sits at the origin. The radar gives the distance d_i to every planet and d_s to the star:

$$|\mathbf{x} - \mathbf{p}_i|^2 = d_i^2, \qquad |\mathbf{x}|^2 = d_s^2. \tag{2}$$

Subtracting the star's equation from each planet's removes $|\mathbf{x}|^2$, the only nonlinear term:

$$2\mathbf{p}_i \cdot \mathbf{x} = |\mathbf{p}_i|^2 + d_s^2 - d_i^2. \tag{3}$$

This gives eight linear equations in two unknowns, solved by least squares in one step, without iteration or an initial guess.

B. Results

The position right after launch is $(0.68814395, 0.00009498)\,\mathrm{AU}$, with all residuals below $0.00\,\mathrm{km}$ (Fig. 6). The radar's errors are proportional to distance. Propagating them through Eq. (3) gives the error map in Fig. 7: about $954\,\mathrm{km}$ per 10^{-6} relative noise where the spacecraft is. Of the noise levels we tested, it first leaves the API's $0.01\,\mathrm{AU}$ tolerance at a relative noise of 1×10^{-2}.

V. MANUAL ORIENTATION

With the three parts combined, the spacecraft finds the following (Fig. 8):

- pointing angle 171.00°;
- velocity $(2.4534, 8.3257)\,\mathrm{AU/yr}$;
- position $(0.68814395, 0.00009498)\,\mathrm{AU}$.

These come from instruments only. Our launch simulation (Parts 1 and 3) predicted a state that agrees to $1.1\,\mathrm{km}$ and $7.4\,\mathrm{m/s}$. `ast2000tools` verified all three (passed). The $7.4\,\mathrm{m/s}$ difference is the API's Euler-integrated launch against our adaptive one. Over a four-month journey it would grow to a large miss, which is why Part 5 re-aims from the measured state.

VI. DISCUSSION AND CONCLUSION

Each instrument has a clear error model:

- the pointing is limited by the contrast of the galactic band;
- the velocity by the spectrograph's precision;
- the position by the radar's relative precision and the planets' geometry.

In all three cases our margins are large. The most important finding was the Doppler convention. Following the assignment's formula would have failed verification with our stars, and only reading the API's source revealed why.

[1] AST2000 teaching team, AST2000 spacecraft project, parts 1–9, Institute of Theoretical Astrophysics, University of Oslo (2023).

[2] K. M. Górski, E. Hivon, A. J. Banday, B. D. Wandelt, F. K. Hansen, M. Reinecke, and M. Bartelmann, HEALPix: A framework for high-resolution discretization and fast analysis of data distributed on the sphere, The Astrophysical Journal **622**, 759 (2005).

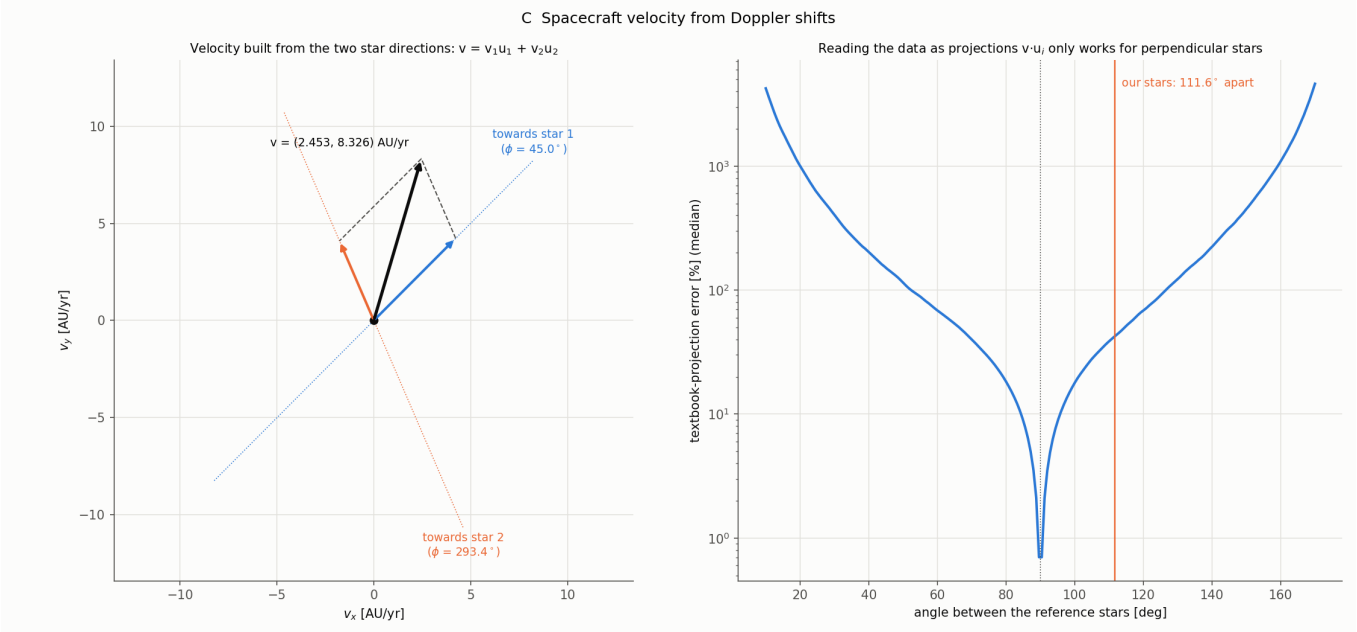


FIG. 4. Left: the velocity built from the two star directions. Right: error of the textbook reading against the angle between the stars.

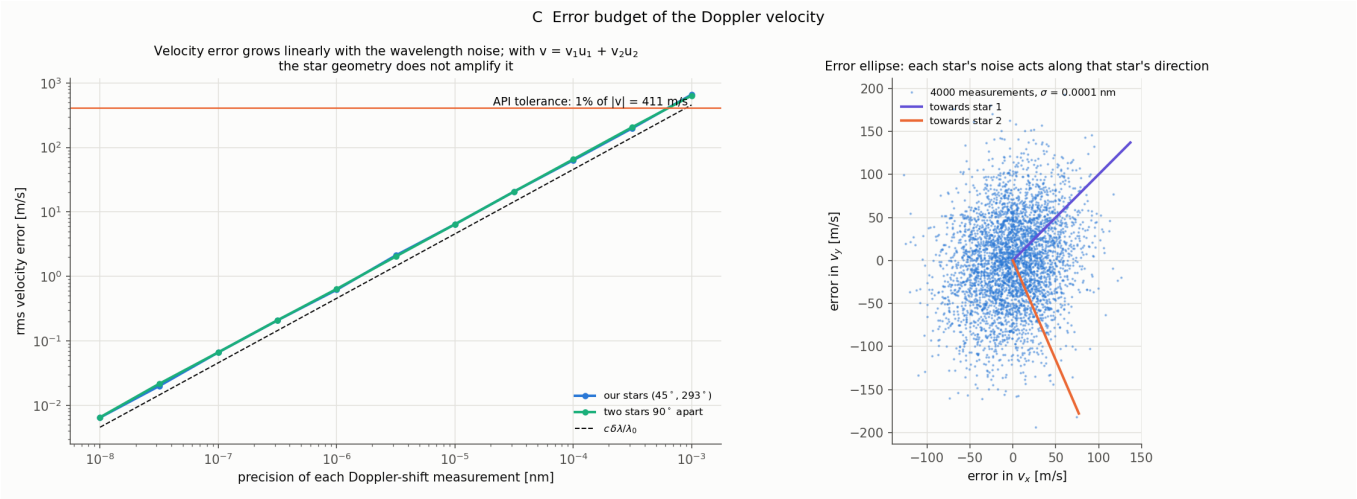


FIG. 5. Left: velocity error against the precision of the Doppler measurement. Right: the error ellipse for 10^{-4} nm noise.

Appendix A: Mission log

Part 4 key numbers (seed 42716)	
Pointing after launch	171.00°
Velocity	(2.4534, 8.3257) AU/yr
Position	(0.68814395, 0.00009498) AU
vs. launch prediction	1.1 km, 7.4 m/s
Worst pointing error (tests)	0.23°
Textbook Doppler error	27%
API verification	passed

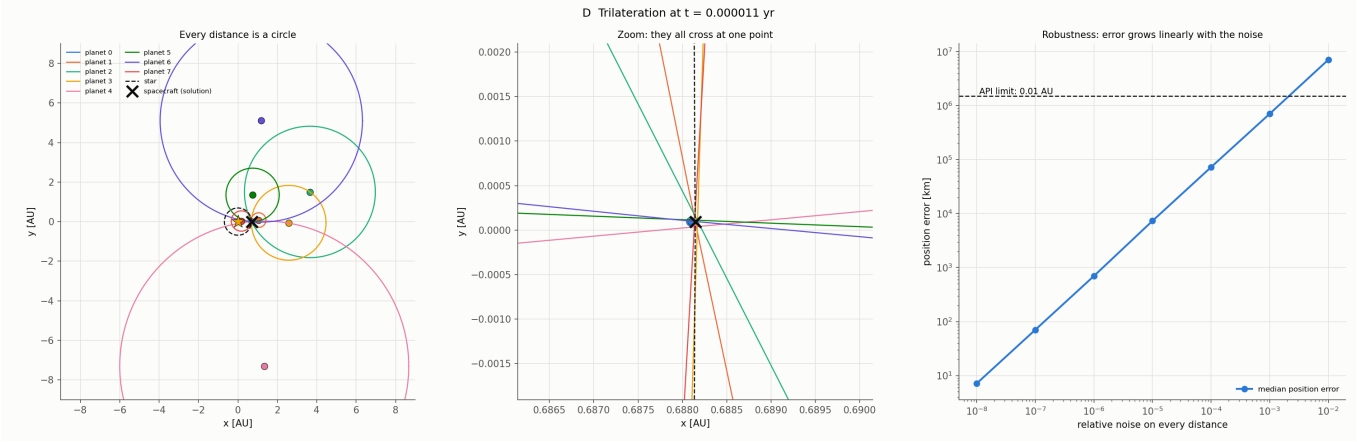


FIG. 6. Every distance is a circle around its planet; all of them cross at the spacecraft. Right: error against noise.

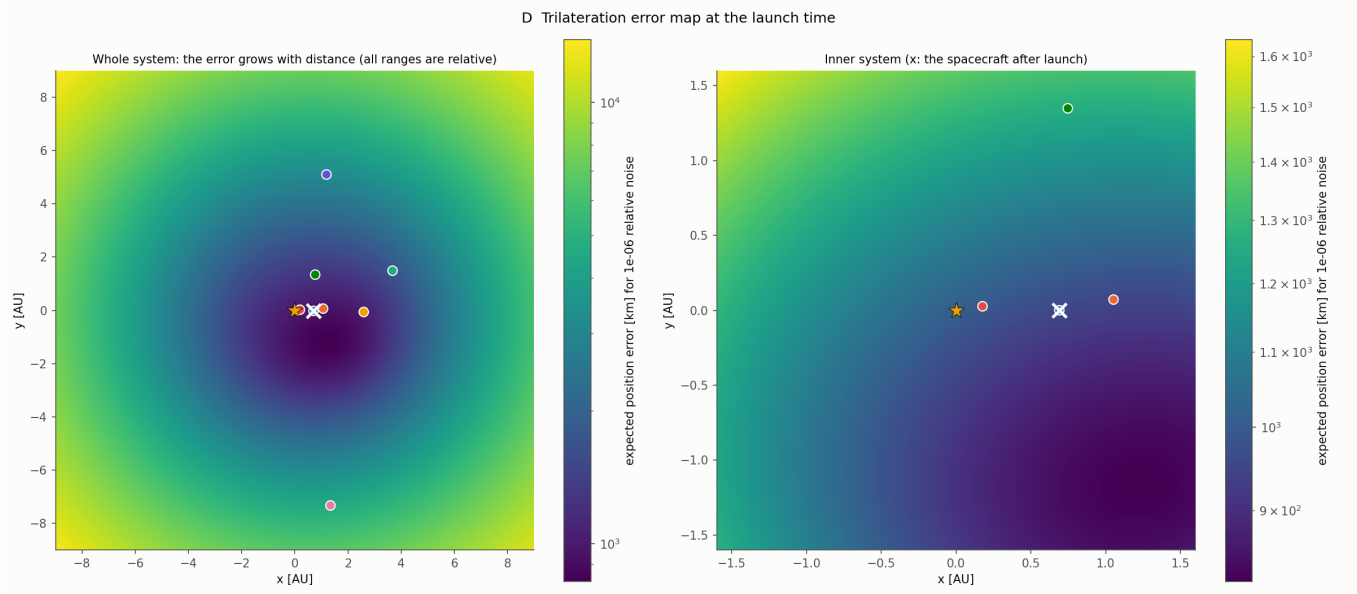


FIG. 7. Expected position error across the system for 10^{-6} relative noise on every distance, from linear error propagation through the least-squares solution.

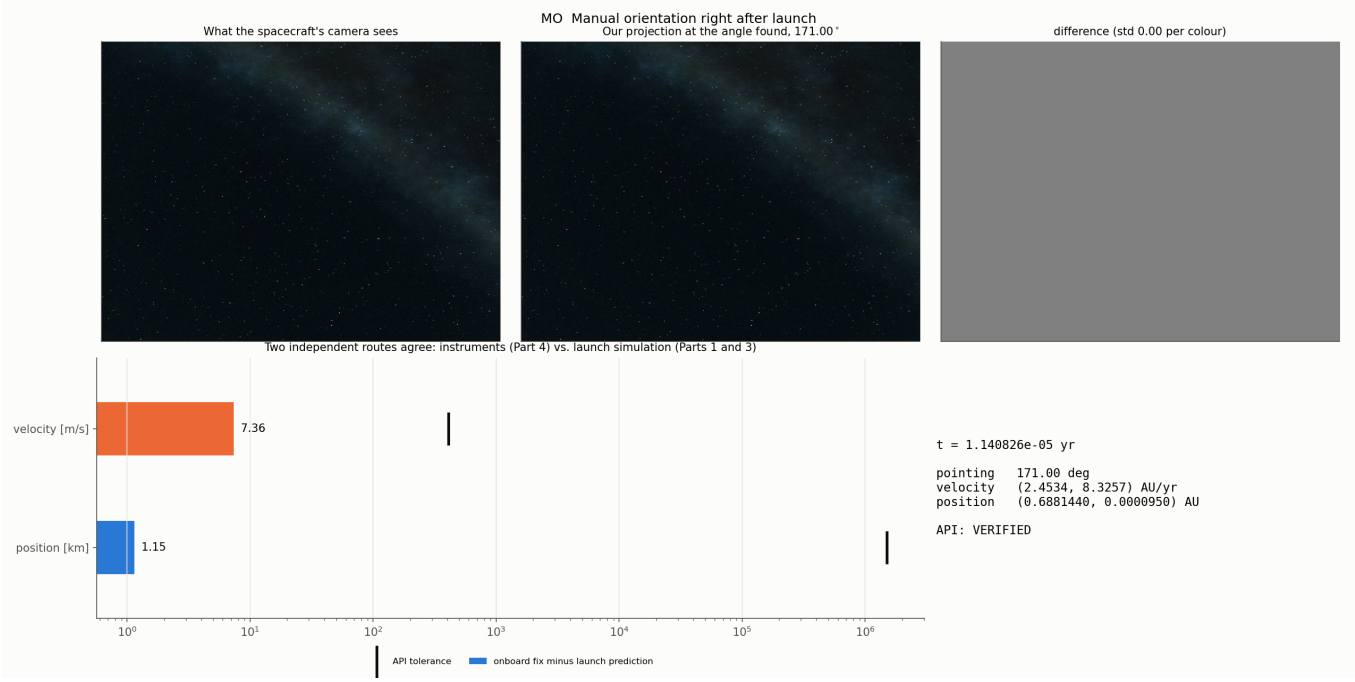


FIG. 8. Manual orientation. The spacecraft's picture, our projection at the angle found, and their difference. Below: onboard fix against launch prediction, with the API's tolerances.